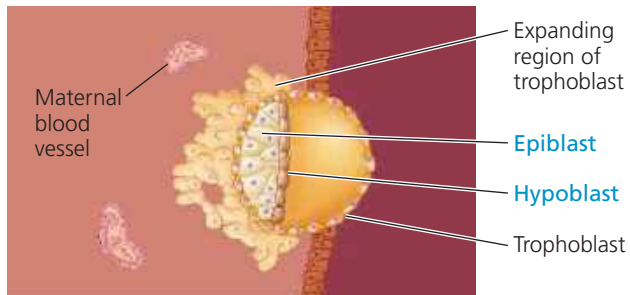
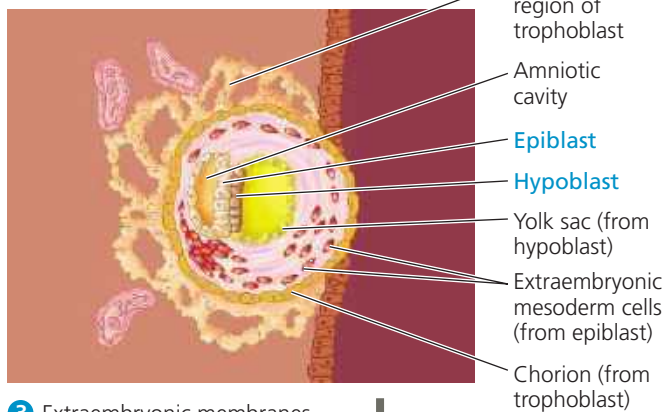


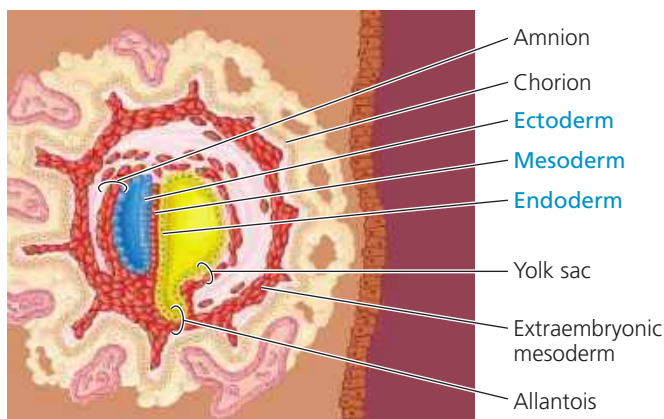
- 1 Blastocyst reaches uterus.



- 2 Blastocyst implants (7 days after fertilization).



- 3 Extraembryonic membranes start to form (10–11 days), and gastrulation begins (13 days).



- 4 Gastrulation has produced a three-layered embryo with four extraembryonic membranes: the amnion, chorion, yolk sac, and allantois.

▲ **Figure 47.12 Four stages in the early embryonic development of a human.** The names of the tissues that develop into the embryo proper are printed in blue.



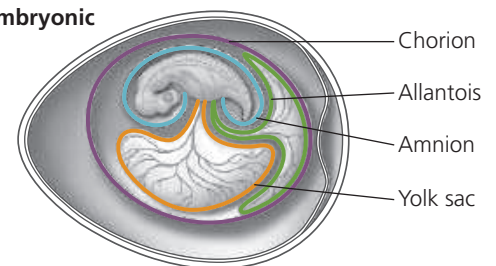
Developmental Adaptations of Amniotes

EVOLUTION During embryonic development, mammals and reptiles (including birds) form four extraembryonic membranes: the chorion, allantois, amnion, and yolk sac (**Figure 47.13**). In all these groups, such membranes provide a “life-support system” for further development. Why did this adaptation appear in the evolutionary history of reptiles and mammals, but not other vertebrates, such as fishes and amphibians? We can formulate a reasonable hypothesis by considering a few basic facts about embryonic development. All vertebrate embryos require an aqueous environment for their development. The embryos of fishes and amphibians usually develop in the surrounding sea or pond and need no specialized water-filled enclosure. However, the extensive colonization of land by vertebrates was possible only after the evolution of structures that would allow reproduction in dry environments. Two such structures exist today: (1) the shelled egg of birds and other reptiles as well as a few mammals (the monotremes) and (2) the uterus of marsupial and eutherian mammals. Inside the shell or uterus, the embryos of these animals are surrounded by fluid within a sac formed by one of the extraembryonic membranes, the amnion. Mammals and reptiles, including birds, are therefore called **amniotes** (see Concept 34.5).

For the most part, the extraembryonic membranes have similar functions in mammals and reptiles, consistent with a common evolutionary origin (see Figure 34.25). The chorion is the site of gas exchange, and the fluid within the amnion physically protects the developing embryo. (This amniotic fluid is released from the vagina when a pregnant woman’s

▼ Figure 47.13 The shelled egg of reptiles.

- (a) The four extraembryonic membranes in a reptile egg



- (b) A baby red-tailed racer snake (*Gonyosoma oxycephala*) hatching from its protective egg